

## 1 PROBLEM STATEMENT Patients cannot understand their Explanation of Benefits

In some countries, healthcare is covered by private insurance companies. After each medical visit, the insurer sends the patient a document called an **Explanation of Benefits (EOB)** — showing what was billed, what the insurer paid, and what the patient owes. Most patients cannot read this document correctly, which leads them to pay bills they do not actually owe.

- **Over-payment:** Patients pay bills the EOB already shows as \$0 owed. Network agreements (Note A0) legally prohibit billing beyond the EOB amount.
- **Unknown deductible status:** Patients cannot easily see how close they are to their annual deductible or out-of-pocket maximum.
- **HRA opacity:** Health Reimbursement Account balances and their interaction with plan payments are invisible to most policyholders.

## 2 TARGET USER Insured patients who received a medical bill or EOB they cannot understand

Any patient with health insurance who received an EOB or a provider bill they cannot reconcile. Key scenarios: patient billed \$160 when the EOB states \$0 owed; patient wanting to know remaining deductible before scheduling further care; patient with an HRA needing to know their current balance and how it was applied to a recent claim. No technical proficiency required — interaction is plain-language Q&A.

## 3 DATA SOURCE Private, patient-specific EOB and benefits documents

- **Private by nature:** EOB documents originate from insurer portals — private and unique to each patient.
- **Patient-specific:** Each EOB carries exact deductible accumulators, HRA balances, claim reference numbers, and provider billing codes unique to that patient and claim.

## 4 AI CAPABILITIES RAG pipeline over unstructured PDFs — every answer grounded in retrieved document chunks

- **Fact:** "What do I owe?" → retrieves the *What I Owe* field from the EOB (\$0).
- **Data:** "How close am I to my deductible?" → retrieves accumulator, computes remainder (\$132.11 of \$1,750 remaining).
- **Combined:** "My plan paid \$0 but I owe nothing — how?" → retrieves HRA payment record across both documents (\$50.82 HRA applied).
- **Actionable:** "Doctor threatening collections" → returns Note A0, reference #9221731700140, Cigna 1-800-244-6224.
- **Citations & confidence:** Source document and section cited per answer. If not found in indexed documents, the chatbot says so — no hallucination.

## 5 TECH STACK Confirmed items locked · [ TBD ] items to be decided and tracked in EPMGDPLAUS-77

| Layer        | Technology / Options                                              |
|--------------|-------------------------------------------------------------------|
| LLM          | Azure AI Foundry   EPAM Dial API Key   Local LLM (oLLama) [ TBD ] |
| Vector store | ChromaDB (local)                                                  |
| Embeddings   | all-MiniLM-L6-v2 via @xenova/transformers (local)                 |
| PDF parsing  | pdf-parse (Node.js)                                               |
| Backend      | Node.js / Express   React / Next.js (API routes) [ TBD ]          |
| Frontend     | Vanilla HTML/JS   React / Next.js chat UI [ TBD ]                 |

LLM preference per EPMGDPLAUS-73: Azure AI Foundry (Azure cloud). Alternatives: EPAM Dial API Key (EPAM infrastructure) | Local LLM via oLLama (worst-case fallback). Backend and frontend framework decisions to precede Week 2 development.

## 6 ARCHITECTURE Application flow — ingestion pipeline and patient query pipeline

